School of Computer Science Engineering and Information Systems

Fall Semester 2025 – 2026

SWE1901 – Technical Answers for Real World Problems

Assignment – 1

Topic: “Food quality Assessing check for image processing / tool base”

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1.Problem Identification:

The problems of quality and safety of food are still prominent concerns of the world, especially in places where food tampering, poor storage, and absence of regulatory frameworks for quality evaluation are common. Enormous quantities of food get consumed on a daily basis without any guarantee of their safety or freshness. Most people depend on the visual inspection, smell, or taste which, in most cases, is insufficient for detecting early spoilage or harmful contamination. Microbial spoilage and some other gas emissions go unnoticed, which may result in severe food poisoning, gastrointestinal illnesses, or chronic diseases in the long run.

In addition to individual health, food quality issues are a hidden driver of economic impact. Households and families of people are wasting their hard-earned money on spoiled food, retailers are suffering heavy financial burdens because of returned stock or damaged reputation and in effect, producers are losing credibility. As the Food and Agriculture Organization estimates, around one third of food produced globally is lost, and undetected spoilage during storage is one major contributor.

The impact we are seeing here stretches far and wide: consumers and their families are facing health risks; they are facing financial burdens from underperforming products; and on a wider scope public, the state of the economy is being impacted.

2. Abstracting the Problem:

To address this widespread issue, we propose developing a cost-effective, sensor-based embedded system that can reliably assess the quality of food using a combination of gas detection, temperature and humidity measurement, and basic image processing. This transforms the real-world issue of food spoilage into a clearly defined technical problem solvable using low-cost, off-the-shelf electronic components.

We assume that indicators such as gas emissions (e.g., ammonia), increased humidity, temperature variation, and visual color change are consistent signs of spoilage. The solution must operate within a budget of ₹1500, be compact, modular, and suitable for use in small-scale environments like homes or small grocery shops.

3. Scientific/Engineering Methodology:

The core hardware includes an Arduino Uno or Nano microcontroller, an MQ-135 gas sensor for detecting volatile organic compounds, and a DHT11/DHT22 sensor for measuring ambient temperature and humidity. Optionally, an ESP32-CAM module will be used to perform image-based color detection, capturing food surface changes during spoilage. The sensors are mounted on a breadboard, connected using jumper wires, and powered using a USB cable or a 9V battery for portability.

The MQ-135 sensor operates on the principle of changing resistance in the presence of gases. DHT sensors use a capacitive humidity sensor and a thermistor to measure environmental conditions. Color analysis via ESP32-CAM is based on HSV (Hue, Saturation, Value) image segmentation, which is more stable under varied lighting than traditional RGB. Alerts will be issued using a buzzer and an LED once threshold levels are surpassed

Software and Tools Used:

- Arduino IDE: Used for writing and uploading firmware to the Arduino and ESP32 boards. It supports libraries like DHT.h, Adafruit_Sensor.h, and ESP32CAM.
- Python (optional): For post-processing logged data (temperature, humidity, gas concentration) and generating graphs or reports.
- OpenCV (with ESP32 or external PC): If advanced image processing is needed, OpenCV can be used to perform more detailed analysis on the color distribution or spoilage patterns.
- Serial Monitor (built into Arduino IDE): For debugging and real-time data reading.
- Processing IDE (optional): To visualize sensor data in real time.
- Tinkercad (for circuit simulation): Used in the initial design phase to simulate sensor behavior before hardware assembly.

4. Design and Development Challenges:

System design and testing will present some difficulties. Sensor integration into a smaller breadboard will likely lead to tangled wiring and signal noise. Additionally, sensor calibration will be crucial, especially that which is related to the gas sensors, and while calibration can be addressed, calibration will be very difficult. Power consumption will need to be minimized for continuous operation; and threshold values will have to be chosen for differential food sources and environmental conditions.

Technically, economically, and socially analytical from a feasibility perspective, all components are compatible and there is ample documentation for each component. The cost to assess completion of materials is reasonable with an approximate investment threshold of ₹1,000–₹1,500. Socially the device will contribute a clear low-cost opportunity to enhance public health and food safety in capacity restricted contexts.

5. Expected Outcomes and Impact:

The system we're building is designed to give early warnings about food spoilage using simple visual and audio signals—like lights and buzzers—so users can act before things go bad. Thanks to the combination of different sensors, the device will monitor food conditions in a more accurate and reliable way. If we include the ESP32-CAM for remote access, the system can also be used in bigger setups like shops, storage units, or during food transport.

The overall impact could be quite significant. It can help reduce food waste, prevent health issues caused by spoiled food, and give people real-time updates about food freshness—all at an affordable cost. Over time, this kind of tech could encourage more people to adopt smart, low-cost solutions for everyday problems. Plus, there's room to upgrade it later with features like machine learning or more advanced IoT functions.

That said, there are still some challenges to keep in mind. The ESP32-CAM has a low-resolution camera, so it might not detect slight changes in food color accurately. Also, since this device may be used in kitchens or storage areas, we need to make sure it's protected with a food-safe, moisture-resistant enclosure to keep the electronics safe. Another concern is usability—without a proper display or mobile app, non-technical users might find it confusing if it only gives feedback through lights and sounds. Lastly, gas sensors need regular maintenance, as they tend to wear out and may need recalibration or replacement over time.

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